

Dynamic Traffic Assignment in Mixed Traffic

Modeling, Evolution, and Control

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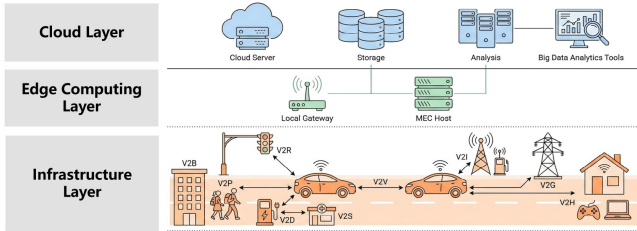
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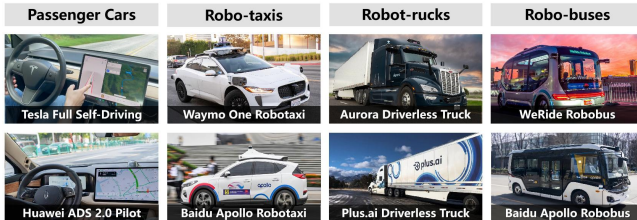
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Emergence of Connected and Autonomous Vehicles in Traffic System

Connected vehicles



Autonomous vehicles



Dynamic Traffic Assignment (DTA)

DTA extends traffic assignment to time-varying conditions.

It describes how **departure times**, **route choices**, and **congestion propagation** jointly determine traffic flows over time, and is used to analyze **dynamic user equilibrium (DUE)** or **system-optimal (DSO)** states under changing demand and supply.

Mixed Traffic with HDVs, AVs and CVs

With the rapid deployment of autonomous vehicles (AVs) and connected vehicles (CVs), road networks are evolving from purely human-driven traffic to mixed fleets.

Vehicles now differ in their automation level (AL) and connectivity level (CL), leading to **heterogeneous** driving behaviour, information availability, and control capabilities in the network.

Key Challenges in Dynamic Traffic Assignment in Mixed Traffic

1. Modeling challenge

Classical DTA with a homogeneous or **binary HDV/CAV** setting cannot capture such **heterogeneous behaviours**, capacities and interactions.

2. Evolutionary challenge

Changes in AL×CL penetration rates, cooperation and information quality can lead to complex **day-to-day dynamics**.
The network may not smoothly converge to an equilibrium, but exhibit **multiple equilibria** or **persistent oscillations**.

3. Control challenge

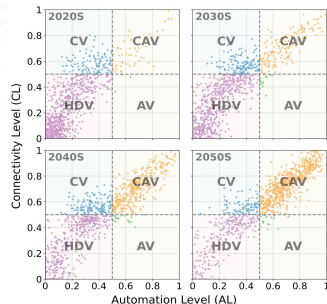
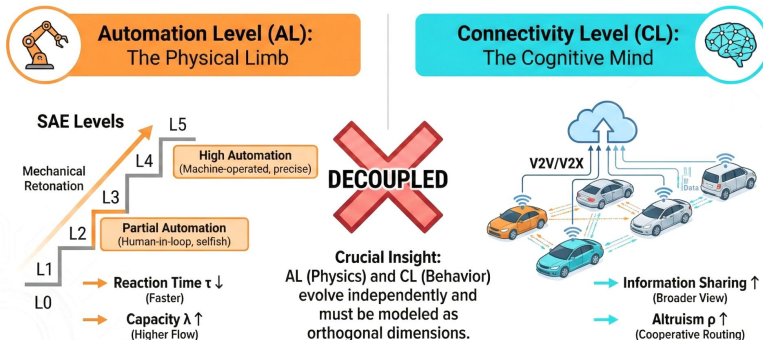
Partial controllability of CAVs creates new levers for **system-level control** (eg. routing, coordination, incentives).
However, it is still unclear how to design strategies that balance **efficiency**, **stability** and **equity** in mixed traffic.

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AL×CL Plane: Decoupling Automation and Connectivity for Mixed Traffic

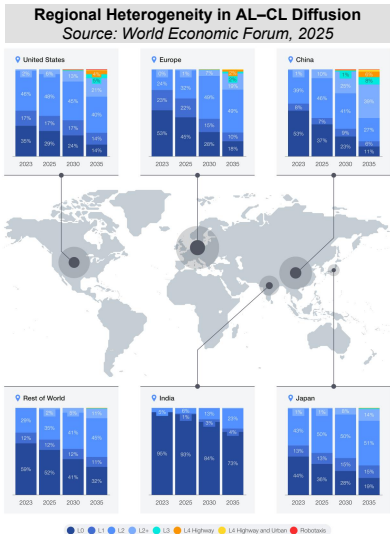
Mixed traffic can be viewed as a time-evolving distribution over the **AL×CL plane**, separating **physical capability** (AL) from **cognitive capability** (CL).

We map each AL×CL state into four interpretable parameters: λ (capacity boost), μ (wave damping) — **physics**; γ (cooperation), ρ (altruism) — **cognitive**.



Example (China): projected evolution on the AL×CL plane (Boston Consulting Group, 2025)

Data-Driven Calibration and Model-Based Evolution of AL–CL Parameters



India

Japan

LD

L1

L2

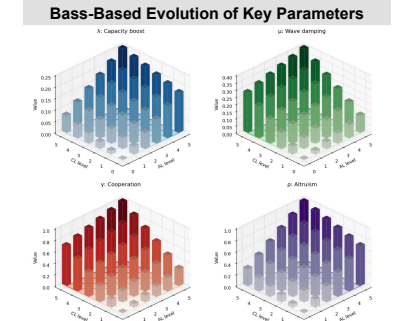
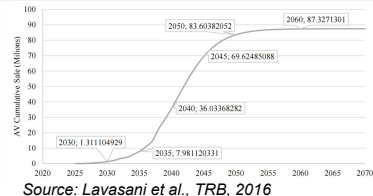
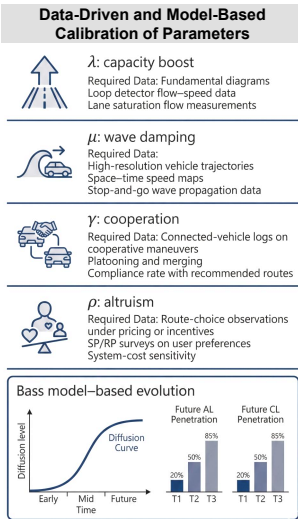
L2+

L3

L4 Highway

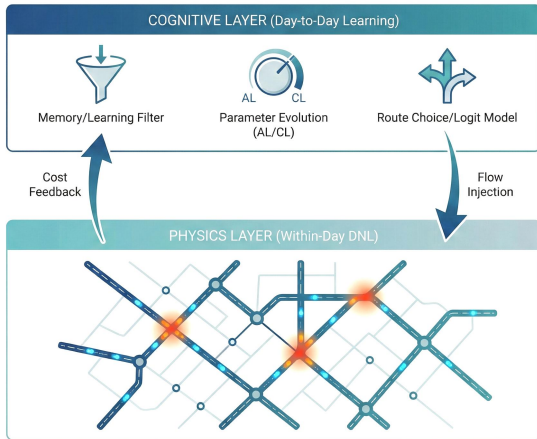
L4 Highway and Urban

Robotaxis



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Dual-Scale Traffic Dynamics: From Within-Day to Day-to-Day



Algorithm 1: Dual-Scale Mixed-Traffic Dynamics on the AL-CL Plane

1 Indices / sets (in order of appearance): day k ; time $t \in [0, T]$; AL-CL state $i \in \mathcal{S}$; event $e \in \mathcal{E}$; OD pair (r, s) ; path $p \in \mathcal{P}_{rs}$.

2 0) AL-CL state $\Rightarrow$ four key parameters

$$(\lambda_i, \mu_i, \gamma_i, \rho_i) = f(AL_i, CL_i), \quad i \in \mathcal{S}. \quad (1)$$

4 1) Event calendar $\Rightarrow$ day- k departure rates

$$\tilde{H}_{rs}^{i,k} = \tilde{H}_{rs}^{i,k} + \sum_{e \in \mathcal{E}} \mathbf{1}_e(k) \Delta_{e,rs}^i + \xi_{rs}^{i,k} \tilde{H}_{rs}^{i,k}, \quad D_{rs}^{i,k}(t) = H_{rs}^{i,k} \varphi_{rs}^i(t). \quad (2)$$

6 2) Within-day DNL as an operator (cost generation)

$$C_p^{i,k}(t) = \left[\mathcal{T}_i \left(D^{i,k}(\cdot), h; \lambda_i, \mu_i, \gamma_i \right) \right]_p(t), \quad i \in \mathcal{S}. \quad (3)$$

8 3) Generalized within-day cost (UE-SO continuum)

$$C_p^{i,k}(t, h) = (1 - \rho_i) C_p^{i,k}(t, h) + \rho_i M_p(t, h), \quad i \in \mathcal{S}. \quad (4)$$

10 4) Within-day equilibrium as a VI (compact presentation form)

11 Find $h^{k*} = \{h_p^{i,k*}(t)\}$ such that for any feasible $\tilde{h}$,

$$\sum_{i \in \mathcal{S}} \sum_p \int_0^T C_p^{i,k}(t, h^{k*}) \left(\tilde{h}_p^i(t) - h_p^{i,k*}(t) \right) dt \geq 0. \quad (5)$$

12 5) Day-to-day learning $\Rightarrow$ route choice (WMA + Logit-ESL)

$$\tilde{G}_p^{i,k+1}(t) = \sum_{n=0}^{N-1} \eta_{i,n} G_p^{i,\exp}(t, h^{k-n}), \quad \eta_{i,n} = \frac{R_i^n}{\sum_{j=0}^{N-1} R_i^j}, \quad (6)$$

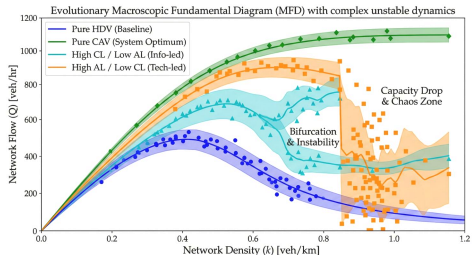
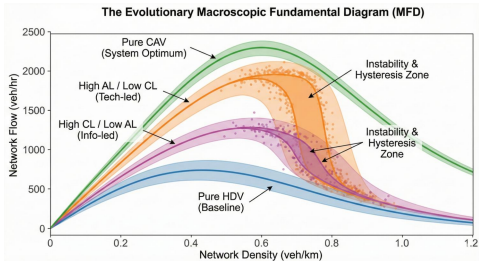
$$\theta_p^{i,k+1}(t) = \frac{\exp[-\beta_i \tilde{G}_p^{i,k+1}(t)]}{\sum_{q \in \mathcal{P}_{rs}} \exp[-\beta_i \tilde{G}_q^{i,k+1}(t)]}, \quad i \in \mathcal{S}. \quad (7)$$

14 6) Evolutionary update (inertia vs. adaptation)

$$h_p^{i,k+1}(t) = (1 - \alpha_i) h_p^{i,k}(t) + \alpha_i \left[D_{rs}^{i,k+1}(t) \theta_p^{i,k+1}(t) \right], \quad i \in \mathcal{S}. \quad (8)$$

$\triangleright \alpha_i \in (0, 1]$ is the learning rate (kept distinct from λ_i).

Nonlinear Traffic Evolution under $AL \times CL$ Heterogeneity



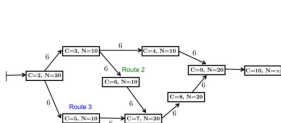
Ideal vs. Reality: Why is that?

Conflicting Behaviors: CVs with different levels of connectivity (UE vs. SO) lead to conflicting traffic behaviors.

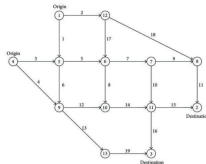
Uneven Distribution: AV penetration is not evenly distributed across the network, with certain areas (e.g., suburban or pilot zones) benefiting less, causing uneven capacity improvements.

Nonlinear Feedback: The interaction between heterogeneous vehicles creates nonlinear feedback effects, where increasing $AL \times CL$ penetration can lead to instability, oscillation, and even chaos in traffic flow.

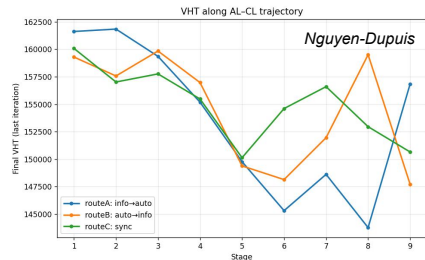
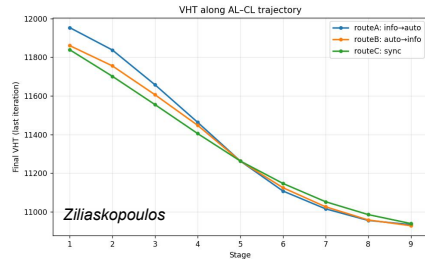
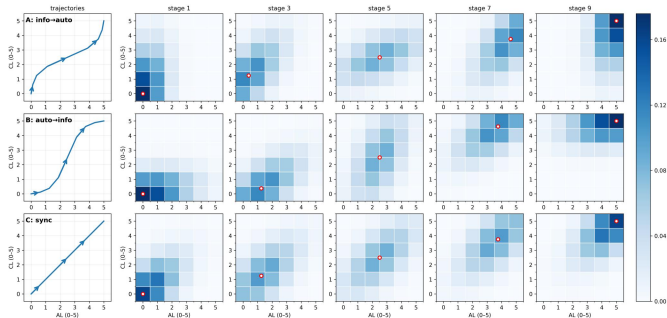
Experimental Insights into Path-Dependent Evolution on the AL×CL Plane



Ziliaskopoulos, 6 links, 7 nodes, 1 OD pair,
Ziliaskopoulos, A. K., 2000



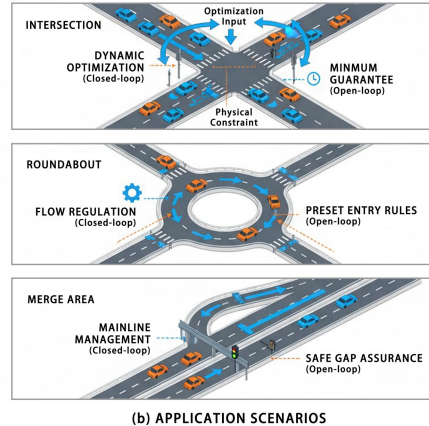
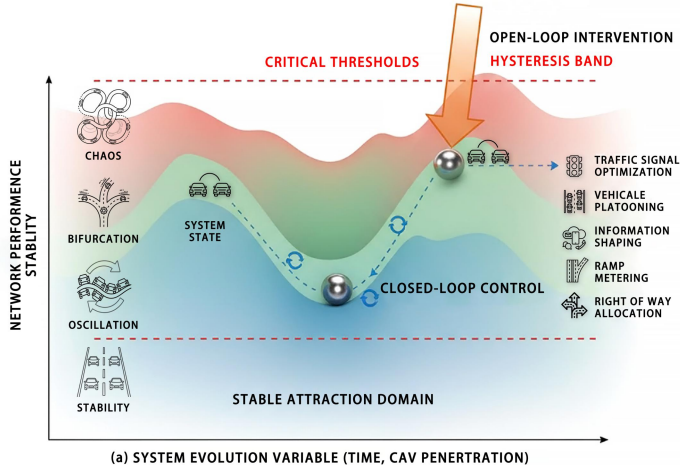
Nguyen-Dupuis, 13 links, 19 nodes, 4 OD
pairs, Nguyen,S.& Dupuis,C, 1984



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Control Strategies for Nonlinear Evolution: Closed-Loop & Open-Loop

Because mixed traffic evolves through nonlinear regimes with **critical thresholds**, effective control must combine **real-time stabilization** (closed-loop) with **boundary shaping** (open-loop).



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Key Findings

→ 1. Modeling: Beyond Penetration Rates

Limitation: Simple binary rates overlook the diversity of Semi-CAVs.

Insight: A decoupled **AL (Automation)** × **CL (Connectivity)** framework provides higher explanatory power.

→ 2. Evolution: Non-Monotonic Transition

Pattern: Efficiency does not improve linearly; the transition phase (20%–60%) is prone to **Braess Paradoxes**.

Risk: Equilibrium existence $\neq$ Stability (Risk of oscillation & chaos).

→ 3. Control: Synergistic Mechanism

Strategy: Adopts a “**Closed-loop Dominance + Open-loop Support**” approach.

Mechanism: Real-time V2X for stability and boundary control for critical thresholds.

Future Outlook

→ 1. Data-Driven Calibration

Move from simulation to empirical evidence. Derive heterogeneous behaviors from real-world trajectory data to calibrate the $AL \times CL$ model.

→ 2. Multi-Objective Equilibrium

Extend optimization goals from Efficiency to **Safety, Equity, and Environment**. Develop mechanisms to guide the system toward Pareto optimality.

→ 3. Phase Transition Management

Quantify **critical thresholds** that trigger phase transitions. Identify boundaries of “Attraction Domains” to prevent traffic chaos.

Thank you for listening!

Q & A session

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